Numerical Analysis Quiz Name: _____

Question 1. Let f be sufficiently smooth and $h > 0$. Consider

$$D_h f(x) = \frac{af(x) + bf(x+h) + cf(x+3h)}{h^2}.$$

Determine the coefficients a, b, and c so that $D_h f(x)$ is a consistent approximation to $f''(x)$. Justify your answer using Taylor expansions about x.

Question 2. Let f be a sufficiently smooth function and let $h > 0$. Consider the following approximation to $f'(x)$:

$$D_h f(x) = \frac{-9f(x-h) + 8f(x) + f(x+3h)}{12h}.$$

Using Taylor expansions with Lagrange remainders, show that there exists $c \in (x-h, x+3h)$ such that

$$D_h f(x) - f'(x) = \frac{1}{2}h^2 f'''(c).$$